

HOMEWORK 5 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

Question Q1. A tank initially contains 2,000 litres of water with a concentration of mercury of 6.0×10^{-5} grams per litre. Water that contains 3.0×10^{-5} grams per litre is pumped into the tank at 100 litres per hour. The well mixed solution is pumped out of the tank at 100 litres per hour. How long will it take for the concentration in the tank to reach 4.5×10^{-5} grams per litre?

Solution

Let $A(t)$ be the amount of mercury (in grams) in the tank at time t (hours). Initial amount

$$A(0) = 2000 \cdot 6.0 \times 10^{-5} = 0.12 \text{ g.}$$

Inflow brings mercury at rate $100 \text{ L/h} \times 3.0 \times 10^{-5} \text{ g/L} = 0.003 \text{ g/h}$. The outflow rate of mercury is $100 \text{ L/h} \times \frac{A(t)}{2000 \text{ L}} = \frac{1}{20} A(t) \text{ (g/h)}$. Hence the ODE is

$$\frac{dA}{dt} = 0.003 - \frac{1}{20} A.$$

This is linear with steady state A_{ss} satisfying $0.003 - \frac{1}{20} A_{ss} = 0$, so $A_{ss} = 0.003 \cdot 20 = 0.06 \text{ g}$. The solution (using integrating factor or standard form) is

$$A(t) = A_{ss} + (A(0) - A_{ss})e^{-t/20} = 0.06 + (0.12 - 0.06)e^{-t/20} = 0.06 + 0.06e^{-t/20}.$$

We want the concentration to be $4.5 \times 10^{-5} \text{ g/L}$, i.e. amount

$$A_{\text{target}} = 2000 \cdot 4.5 \times 10^{-5} = 0.09 \text{ g.}$$

Solve $0.09 = 0.06 + 0.06e^{-t/20}$:

$$0.03 = 0.06e^{-t/20} \quad \Rightarrow \quad e^{-t/20} = \frac{1}{2}.$$

Hence

$$-\frac{t}{20} = \ln \frac{1}{2} \quad \Rightarrow \quad t = 20 \ln 2 \approx 13.86 \text{ hours.}$$

Therefore it will take $t = 20 \ln 2 \approx 13.9$ hours for the concentration to reach $4.5 \times 10^{-5} \text{ g/L}$.

Question Q2. A vat with capacity of 500 gallons initially has 10 lbs. of salt dissolved in 100 gallons of brine. A brine solution with 0.5 lbs. of salt per gallon is pumped into the tank at a rate of 4 gallons per minute, and the well mixed solution is pumped out at a rate of 2 gallons per minute. How much salt will be in the vat when the vat is full? (Hint: first find the solution for the volume of the fluid in the tank, before tackling the amount of salt in it.)

Solution

Let $V(t)$ be the volume (gallons) in the tank at time t (minutes). With inflow 4 gal/min and outflow 2 gal/min,

$$V(t) = 100 + (4 - 2)t = 100 + 2t.$$

The tank fills (reaches 500 gal) at time t_f with $100 + 2t_f = 500$, so $t_f = 200$ minutes.

Let $S(t)$ be the amount of salt (lbs) in the tank at time t . The inflow brings salt at rate $4 \text{ gal/min} \times 0.5 \text{ lb/gal} = 2 \text{ lb/min}$. The outflow removes salt at rate $2 \text{ gal/min} \times \frac{S(t)}{V(t)} = \frac{2S(t)}{100 + 2t}$ (lb/min). Thus

$$\frac{dS}{dt} = 2 - \frac{2S}{100 + 2t}.$$

Simplify the coefficient: $\frac{2}{100 + 2t} = \frac{1}{50 + t}$, so the ODE becomes

$$\frac{dS}{dt} + \frac{1}{50 + t}S = 2.$$

Integrating factor $\mu(t) = \exp\left(\int \frac{1}{50+t} dt\right) = 50 + t$. Multiply through:

$$\frac{d}{dt}[(50 + t)S] = 2(50 + t).$$

Integrate:

$$(50 + t)S = \int 2(50 + t) dt = 100t + t^2 + C.$$

Use $S(0) = 10$ to get $50 \cdot 10 = C$, so $C = 500$. Therefore

$$S(t) = \frac{t^2 + 100t + 500}{50 + t}.$$

When the vat is full at $t = 200$ minutes,

$$S(200) = \frac{200^2 + 100 \cdot 200 + 500}{50 + 200} = \frac{40000 + 20000 + 500}{250} = \frac{60500}{250} = 242 \text{ lbs.}$$

Thus the vat contains 242 lbs of salt when full (after 200 minutes).

Question Q3. A car's fuel tank holds 15 gallons of gasoline. When the tank is full, the gasoline contains 0.015 gallons of ethanol. Suppose that pure gasoline is pumped into the tank at a rate of 1 gallon per minute, and the well mixed solution is pumped out at a rate of 1 gallon per minute. How long will it take for the amount of ethanol in the tank to be 0.003 gallons?

Solution

Let $E(t)$ be the amount (gallons) of ethanol in the tank at time t (minutes). Volume is constant at 15 gallons. Pure gasoline (no ethanol) flows in, so inflow adds no ethanol.

Outflow removes ethanol at rate $1 \text{ gal/min} \times \frac{E(t)}{15} = \frac{1}{15}E(t)$. Thus

$$\frac{dE}{dt} = -\frac{1}{15}E.$$

Solve: $E(t) = E(0)e^{-t/15}$ with $E(0) = 0.015$. We want $E(t) = 0.003$, so

$$0.003 = 0.015e^{-t/15} \quad \Rightarrow \quad e^{-t/15} = \frac{0.003}{0.015} = 0.2.$$

Hence

$$-\frac{t}{15} = \ln(0.2) \quad \Rightarrow \quad t = -15 \ln(0.2) = 15 \ln 5 \approx 24.14 \text{ minutes.}$$

Therefore it takes $t = 15 \ln 5 \approx 24.1$ minutes for the ethanol amount to drop to 0.003 gallons.